



FACULTÉ
D'INFORMATIQUE

CHAPTER 1

INTRODUCTION TO DATABASE

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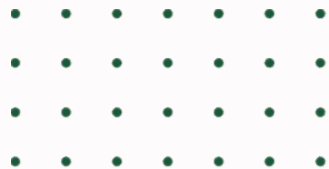
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References

01

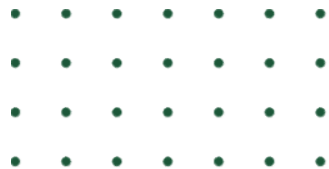
Database systems: Volume 1.
D. Lewis.

<https://www.london.ac.uk/sites/default/files/study-guides/database-systems.pdf>

02

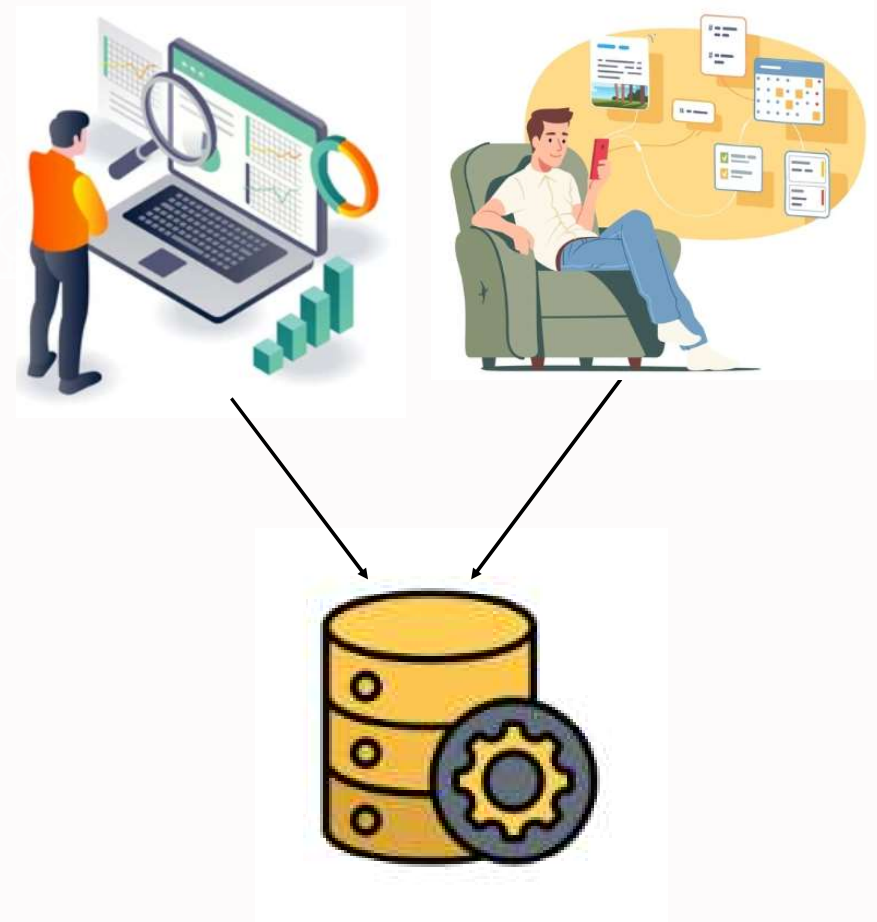
DATABASE SYSTEM CONCEPTS.
Abraham Silberschatz.

<https://people.vts.su.ac.rs/~peti/Baze%20podataka/Literatura/Silberschatz-Database%20System%20Concepts%206th%20ed.pdf>



What is a database ?

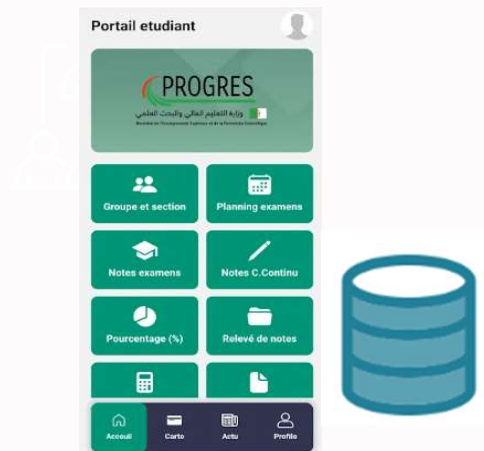
- A Database is a **shared** collection of **related** data designed to meet the information needs of an organization.
- The purpose of a database is to **store** information and to allow users to **retrieve and update** that information on demand.



Examples: Databases in daily life

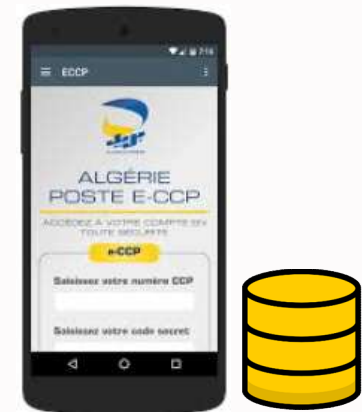
University Database: Progres

Students access their scholar profile, marks.



ECCP

Users access post office services and check their credit/debit..



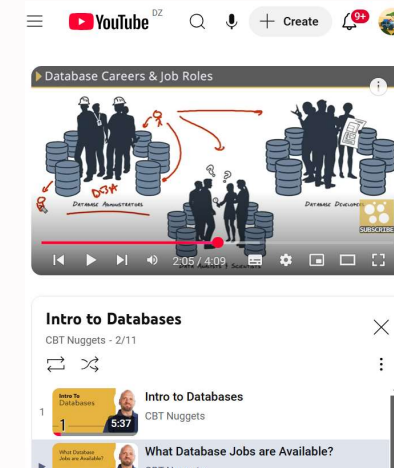
Facebook

Users publish messages, post comments, and send notifications, etc.



Youtube

Users publish and share videos, post comments, and send notifications.



Before Databases...

Manual approach



- Cards and paper are used for handling information.
- Files for as many event and objects as the organization has are used to store information.
- Each of the files containing various kinds of information are classified and stored in different cabinet.
- Files are kept in safe places for security purpose.
- Search for information is done by searching first for the right cabinet then for the right the file

Limitations ?

- Can't control errors.
- Limited to small size information
- Difficult to update, retrieve information.
- Difficult to share data.
- Difficult to control access.
- Difficult to recover if lost.

Before Databases...

File based systems



- Computerize the manual filing system.
- A File is a collection of records which contains logically related data.
- file-based system is a collection of application programs, each managing its own data.

Limitations ?

- Duplication or redundancy of data.
- Limited data sharing.
- Separation and isolation: an application has access to its own files, but no access to the files of the other applications.
- Program-Data dependence
- Limited query processing.

Sales Department files

Property for rent file								
Property ID	Street	Area	City	Postcode	Type	No Of Rooms	Rent	Owner ID

Owner ID				
Owner ID	First Name	Last Name	Address	Telephone Number

Renter file							
Renter ID	First Name	Last Name	Address	Telephone Number	Preferred type	Maximum Rent	

Contracts Department files

Property for rent file					
Property ID	Street	Area	City	Postcode	Rent

Renter file				
Renter ID	First Name	Last Name	Address	Telephone Number

Owner file				
Owner ID	First Name	Last Name	Address	Telephone Number

Lease file									
Lease ID	Property ID	Renter ID	Owner ID	Payment Type	Rent	Deposit	Start	Finish	

Example: A file-based system for an estate agent's company.

Database Management System (DBMS)

- A database management system is a software system that provides a set of primitives for defining, accessing and maintaining a database.
- DBMS provides a multi user access to create, update, store, retrieve data in a database.
- DBMS controls data access, enforces data integrity, manages concurrency control, and recovery.



DBMS Examples:
MySQL, Postgres,
Oracle, SQL
Server.

Database Management System

Objectives

DBMS covers the limitation of File based systems by offering:

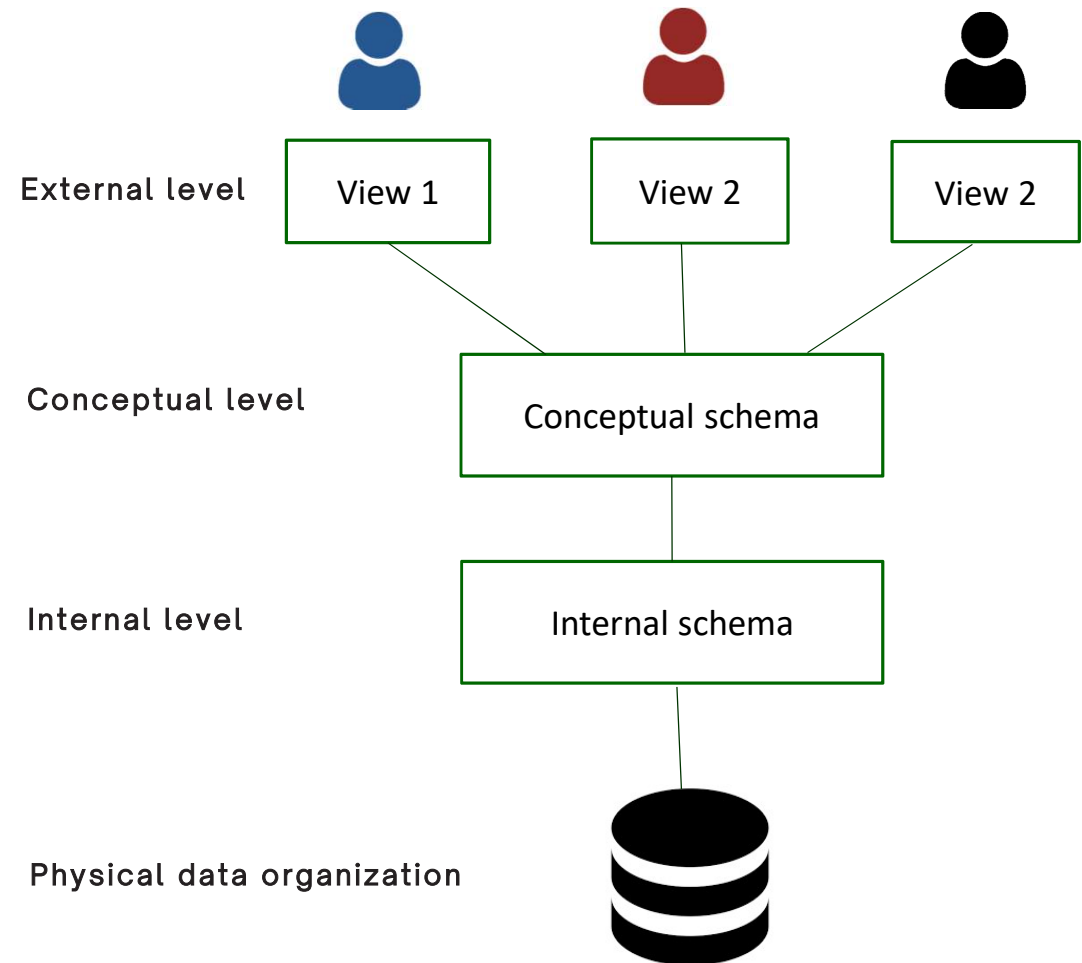
- Data integrity** define constraints and rules to ensure that data is consistent and accurate.
- Data security** define access control rules.
- No redundancy** reduce update operation and having coherent data.
- Accessibility** multiple users can access the data simultaneously, ensuring data availability and coherence.
- Data Sharing** different application can access the same data.
- Data integration** integrating data from different sources.
- Recovery** ensure data is not lost in case of a system failure.
- Data independance** separates the logical and physical view of data and allows users to have different views of the data.

Database Management System

Three level Architecture

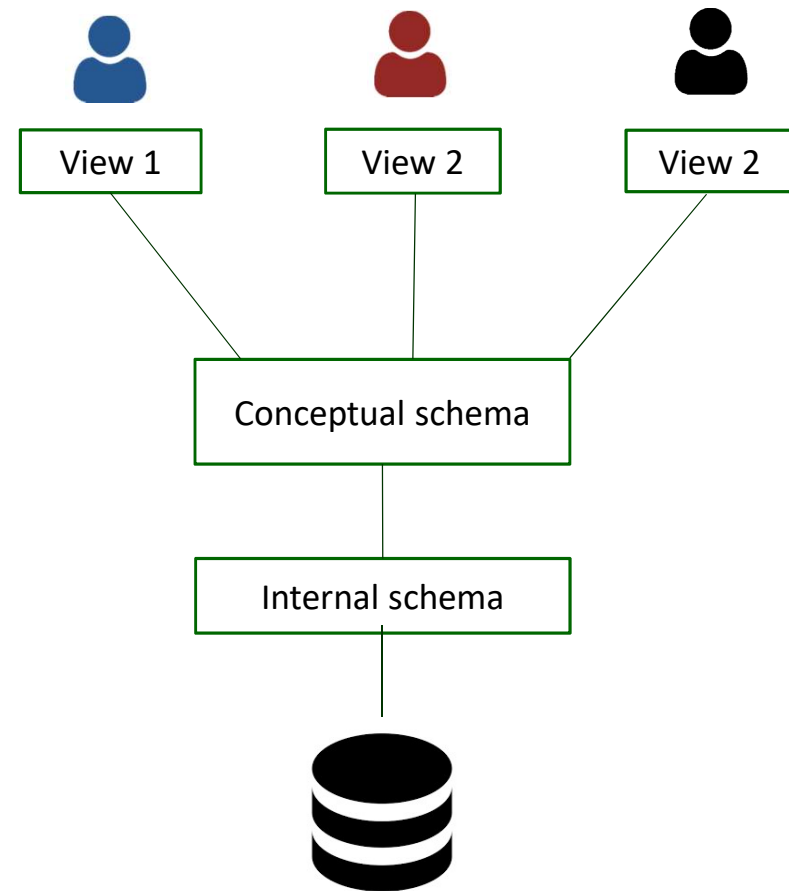
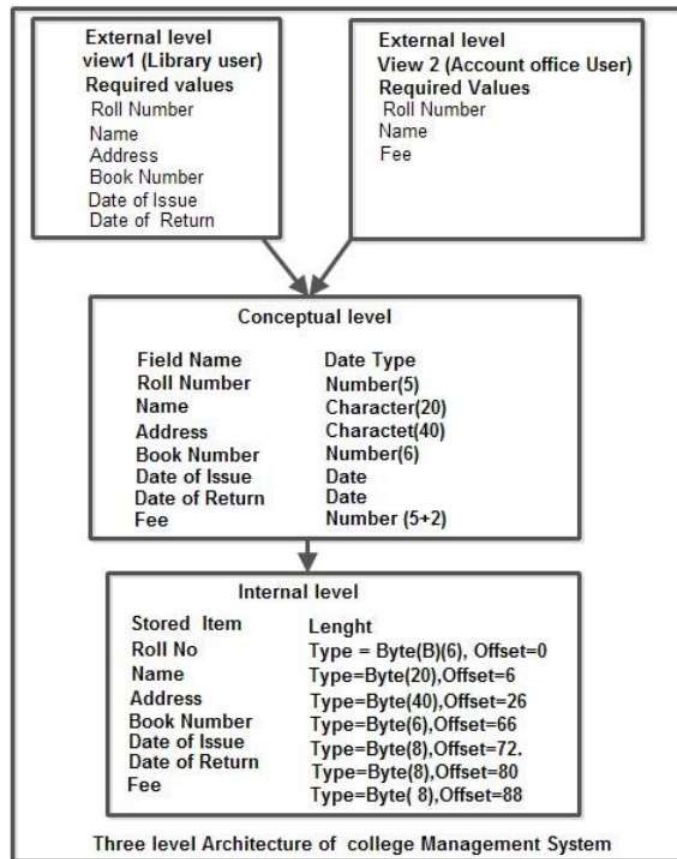
ANSI-SPARC Architecture standard 1975

- Goal: Program Data independence.
- Separation of the design of a database from the details of its implementation.
- Changing the physical representation of data without affecting users, as long as its logical structure is preserved
- Each user needs to have a customized view of the database, not all the database.



Database Management System

Three level Architecture - Example



Database Models

- Physical data independence is one of the main advantages of database systems. This is achieved based on the conceptual level.
- In conceptual level, data is described in terms of its characteristics– its logical structure -> Data model.
- A database model shows the logical structure of a database, including the relationships and constraints.
- There are many kinds of data models. Some of the most common ones include:
 - Hierarchical database model
 - Network model
 - Entity-relationship model
 - Relational model
 - Object-oriented database model
 - The object-relational model
 - Document model, etc.

Database Languages

01 Data Definition Language (DDL)

Allows to define and create objects of the database.

02 Data Manipulation Language (DML)

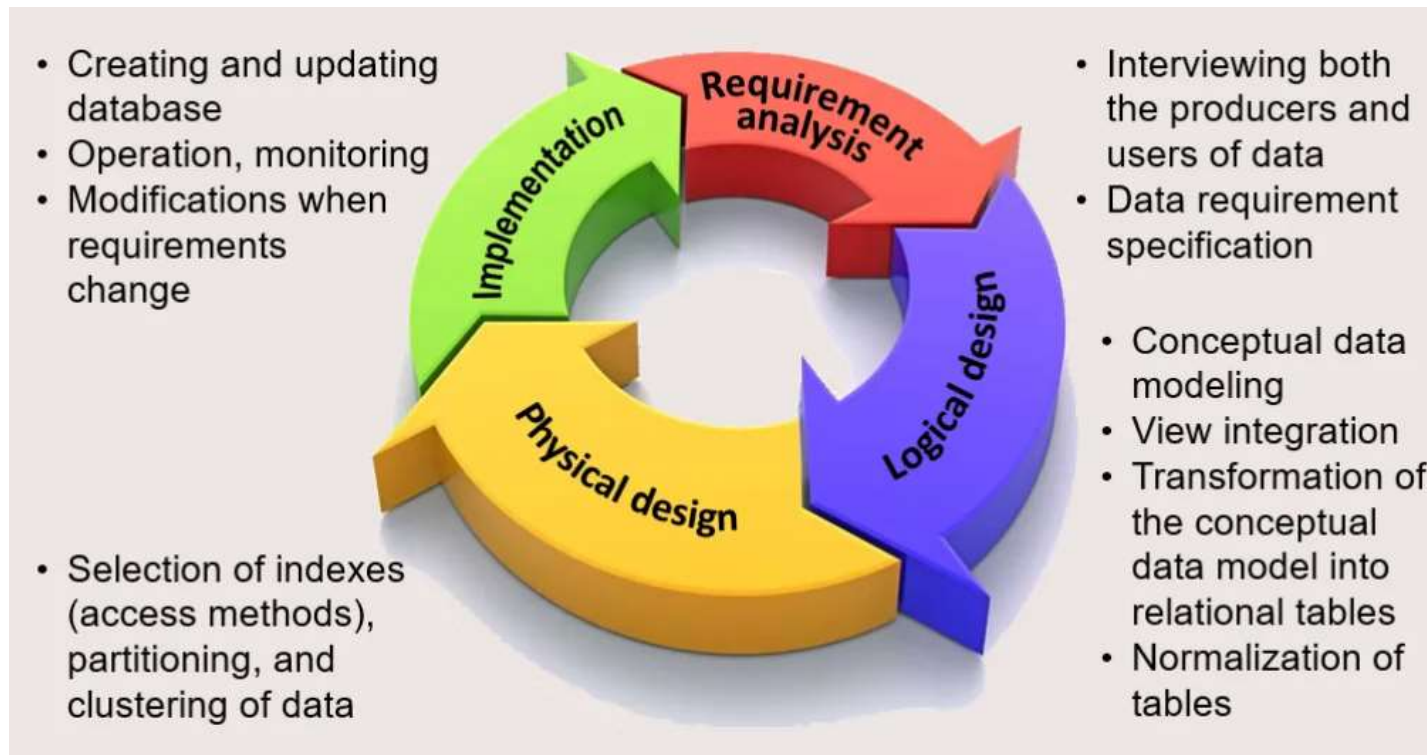
Provides basic data manipulation operations on data held in the database:
Retrieve, insert, update, delete.

Example:

SQL (Structured Query Language)

for relational database.

Database development life cycle



Source: <https://medium.com/informatics/the-database-life-cycle-4151969ca57d>

Database users

01 Data Administrator

- Responsible on management of data resources.
- Involves in database planning.
- Understands the data requirements of the organization
- Decides which data is relevant and which is not.

02 Database Designer

- Is responsible on management of data resources.
- Involves in database planning.
- Understands the data requirements of the organization
- Decides which data is relevant and which is not.

03 Database Administrator

- Responsible for the physical realization of the database.
- Responsible to control and manage the database resources
- Deals with both the correctness of the implementation and the efficiency of the database system.
- Authorizing access to the database.

04 Application Programmer

- Writes programs that perform processing of data of the database.
- Determines the interface that help to retrieve, insert, update and delete data in the database.

05 End users

Users who requires accessing the database frequently for various purpose. We cite 2 groups of users in this category:

Naïve users

- interact with the system using application programs.
- Is not aware of the functionality of the DBMS.
- All they need is reliable and easy access to information.

Sophisticated users

- Are users familiar with the structure of the Database and facilities of the DBMS.
- Access the database directly, through the database language supported by the DBMS.
- Are most of the time engineers, scientists, business analysts, etc.



Next Database Models